F79MB Coursework: Assessed Project 1

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1. a) Numerical summary of the time required to resolve reported software bugs in the United States for the year 2022:

```
 Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
29.00   69.75   95.00  108.55  123.50  560.00
```

Sample standard deviation is 68.12 minutes.

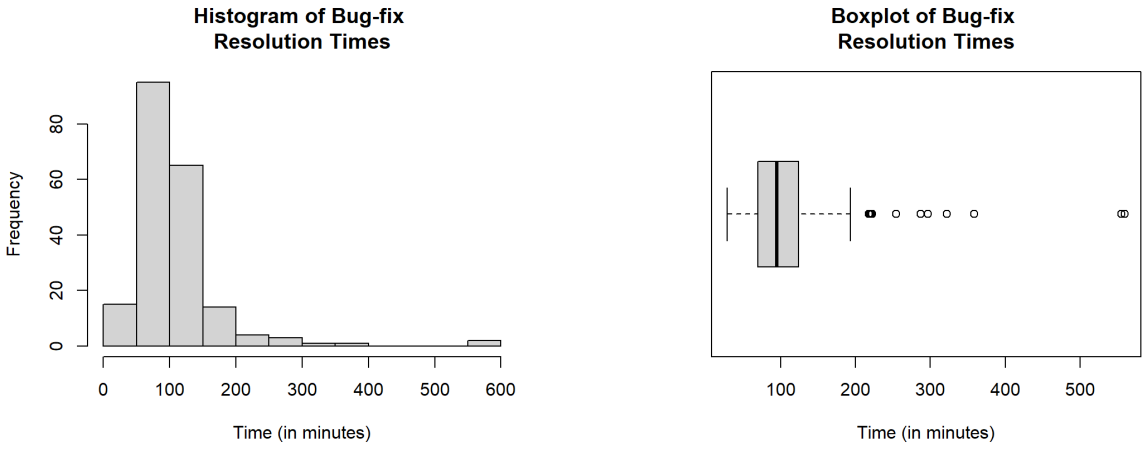


Figure 1: Histogram and boxplot of bug-fix resolution time (in minutes) reported in the United States in 2022, based on a sample of 200 reported software bugs

Comment: From the histogram in Figure 1, the distribution seems to be highly right-skewed and also unimodal, with the highest resolution time at 560 minutes. Numerical summaries also show that mean is larger than the median. The boxplot indicates the presence of some outliers at the upper end of the distribution. The minimum resolution time is 29 minutes and the standard deviation is 68.12 minutes.

b) i) Let $x_1, x_2, \ldots, x_n$ be a random sample from a normal distribution $N(\mu, \sigma^2)$ representing the log-transformed bug-fix resolution times. Then, the probability density function (PDF) for a single observation x_i is:

$$f(x_i|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

The likelihood function $L(\mu, \sigma^2)$ is:

$$L(\mu, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

$$\implies L(\mu, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2\right)$$

Taking the natural logarithm $l = \ln L$:

$$l(\mu, \sigma^2) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2$$

Differentiate l with respect to μ and set it to zero:

$$\frac{\partial l}{\partial \mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i - \mu) = 0$$

$$\therefore \sum_{i=1}^{n} x_i - n\hat{\mu} = 0 \implies \hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

Differentiate l with respect to σ^2 and set it to zero:

$$\frac{\partial l}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}\sum_{i=1}^{n}(x_i - \mu)^2 = 0$$

Multiply by $2(\sigma^2)^2$:

$$-n\hat{\sigma}^2 + \sum_{i=1}^{n}(x_i - \hat{\mu})^2 = 0$$

$$\therefore \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \hat{\mu})^2$$

The Maximum Likelihood Estimators are:

- $\hat{\mu} = \bar{x}$
- $\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \hat{\mu})^2$

b) ii) Using the given data, the computed numerical MLEs of μ and σ^2 are 4.563369 and 0.2187059 respectively. The sample mean and sample variance of the log-transformed data are 4.563369 and 0.2198049 respectively. $\hat{\mu}$ is exactly the same as sample mean as it is derived to be so. However, the sample variance is slightly higher than $\hat{\sigma}^2$. This indicates that using $\hat{\sigma}^2$ accounts for slightly less variation in the data due to $\hat{\sigma}^2$ having n as denominator compared to sample variance that has $n - 1$ as denominator.

c) The Q-Q plot for the log-transformed data as shown in Figure 2 appears to be closely normally distributed. However, a few points on the upper tail lie above the straight line, and a few points on the lower tail lie slightly above the straight line. The histogram and boxplot in Figure 3 further support that the data deviates slightly from the normal distribution for the log-transformed data. Both plots show slight right-skewness in the log-transformed data.

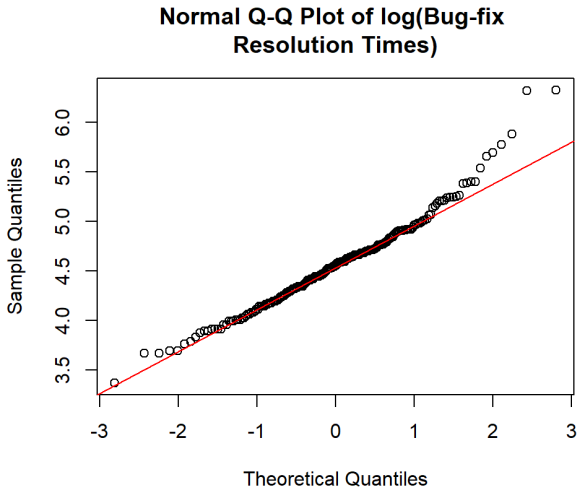


Figure 2: Q-Q plot for log-transformed data.

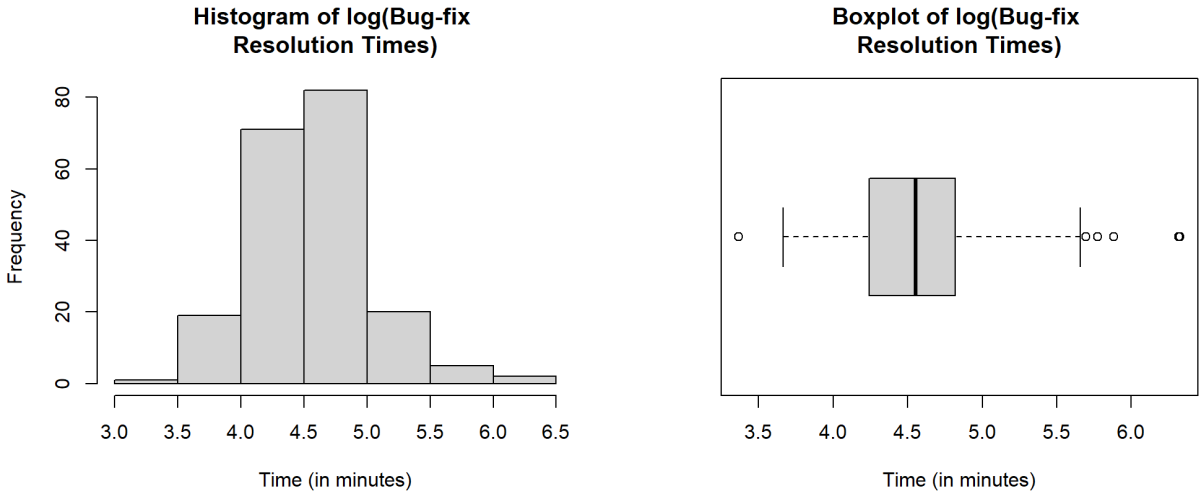


Figure 3: Histogram and boxplot for log-transformed data.

d) Performing a χ^2 goodness-of-fit

H_0 : The log-transformed data has a Normal distribution

H_1 : The log-transformed data does not have a Normal distribution

Let the original data be $x_1, x_2, \ldots, x_n$. we estimate from the data using the MLE estimator as calculated in b(ii), $\hat{\mu} = \bar{x} = 4.563369$ and $\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \hat{\mu})^2 = 0.2187059$ since the null hypothesis does not specify the parameters of $\hat{\mu}$ and $\hat{\sigma}^2$ of a typical Normal distribution. Expected frequencies $e_i = \frac{200}{10} = 20, i = 1, 2, \ldots, 10$ are given from a χ^2 test with 10 equal-probability cells. The results computed in R are summarised in the following table:

Interval (resolution time)	**Observed frequency,** f_i
$[0.00, 52.66963)$	17
$[52.66963, 64.70101)$	22
$[64.70101, 75.04803)$	22
$[75.04803, 85.19043)$	23
$[85.19043, 95.90608)$	17
$[95.90608, 107.96960)$	24
$[107.96960, 122.56121)$	24
$[122.56121, 142.16126)$	19
$[142.16126, 174.63532)$	12
$[174.63532, \infty)$	20

The test statistic value is $\chi^2 = \sum_{i=1}^{10} \frac{(f_i - e_i)^2}{e_i} = 6.6$. Since 2 parameters of normal distribution were estimated, the test statistic χ^2 distribution has $df = 10 - 2 - 1 = 7$, so the p-value $= P(\chi^2_7 > 6.6) = 0.4716799$. Thus, we fail to reject H_0. Overall, it appears that the log-transformed data fits a Normal distribution quite well. It is worth noting that while the Q-Q plot in Figure 2 does not seem to follow a normal distribution exactly, and the diagrams in Figure 3 look a bit right-skewed, the χ^2 goodness-of-fit test points that there is no evidence to reject the null hypothesis that the log-transformed data has a Normal distribution.

2. a) Numerical summary of the duration of user sessions (in minutes) recorded for a mobile application in the United Kingdom during the year 2025:

```
Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
3.00    9.00   15.50   16.62   22.50   40.00
```

Sample standard deviation is 9.33 minutes.

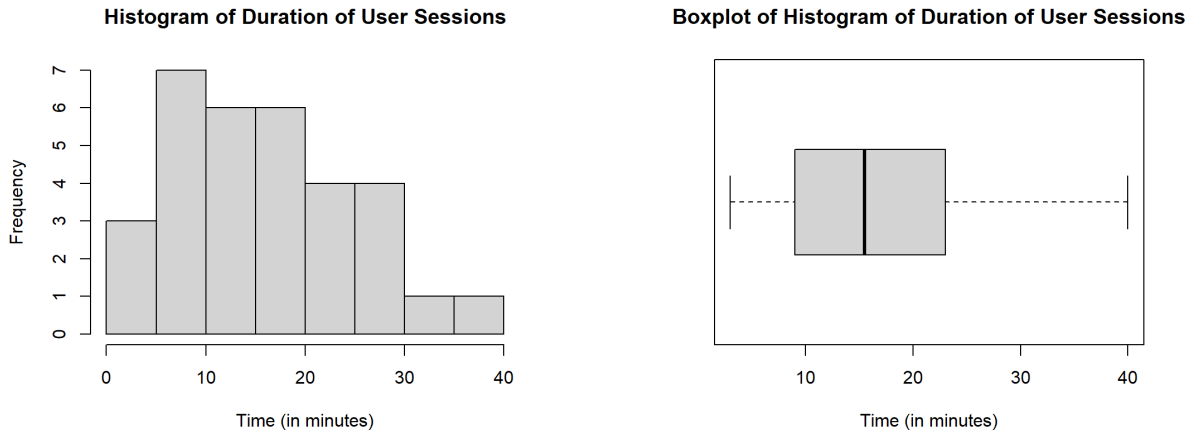


Figure 4: Histogram and boxplot of duration of user sessions (in minutes) recorded for a mobile application in the United Kingdom in 2025, based on a sample of 32 sessions

Comment: From the histogram (Figure 4), the distribution appears to be unimodal and highly right-skewed, with the highest user session duration at 40 minutes. Numerical summaries also indicate that the mean is slightly higher than the median. The minimum user session duration is 3 minutes while the standard deviation is 9.33 minutes. The boxplot also shows right-skewness with no outliers whatsoever.

Choice of statistical methods for inference: non-parametric bootstrap methodology. This is due to the small sample size of 32. Also, the right-skewed nature of the dataset voids normality assumptions that are a critical assumption to fulfill in many other classical methods.

b) The empirical bootstrap sampling distribution of the sample median, $\hat{m}$, is shown in Figure 5. The histogram and the box-plot show a slightly positively skewed distribution. However, the numerical summary indicates that the mean is slightly smaller than the median but somewhat similar.

Bootstrap 95% Confidence Interval for the population median user session duration is (10.99, 19.51).

```
Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
9.00   14.00   15.50   15.45   17.00   23.00
```

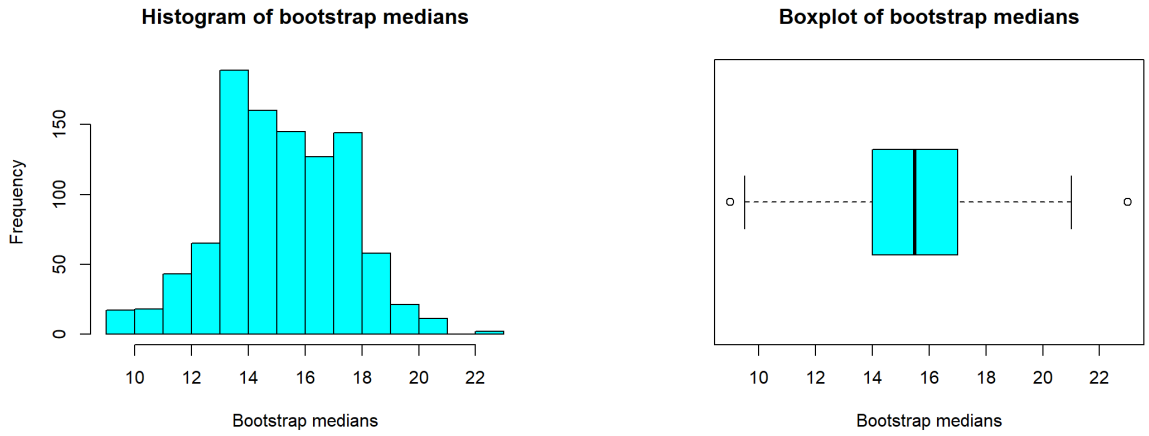


Figure 5: Distribution of bootstrap medians

c) Assuming that the app session duration data can be reasonably modeled by a lognormal distribution with parameters μ and σ^2, we find the estimates from the data to be $\hat{\mu} = 2.632029$ and $\hat{\sigma}^2 = 0.407547$ using the formula derived above in 1. b(i).

Under the asymptotic theory, we should have $\hat{m} \sim N(m, \left(\frac{1}{2f(m)\sqrt{n}}\right)^2)$, where $f(\hat{m})$ is the density of $N(\hat{\mu}, \hat{\sigma}^2)$ distribution and $\hat{m}$ is the log-transformed sample median. We have $\hat{m} = 2.740319$ and sample size $n = 32$. From R, we obtain $f(\hat{m}) = 0.6249154$. The estimate of the standard error of median is approximately $se(\hat{m}) = \frac{1}{2f(m)\sqrt{n}} = 0.1414405$.

Then a 95% confidence interval of the log (user sessions duration), $\hat{m}$, is $\hat{m} \pm se(\hat{m}) = (2.463, 3.018)$.

Taking the exponent of the confidence interval above would result in the 95% confidence interval of the user sessions duration, i.e., (11.741 20.441).

By employing the non-parametric bootstrap and asymptotic theory respectively, the 95% confidence intervals obtained in parts b and c are slightly different but are still relatively close to one another. This is because the sampling distribution of the bootstrap medians in Figure 5 is only very slightly right-skewed, and quite close to a normal distribution, as can be seen from both the Q-Q plot and the numerical summary of the data. With a sample size of $n = 32$, the asymptotic theory is expected to not perform well, since this is quite a small sample size. Additionally, as shown in the histogram and boxplot in Figure 6 below, the log-transformed data being left-skewed raises concerns regarding the normality assumption required for the asymptotic theory to work. While both confidence intervals are quite similar, of the two confidence intervals, the non-parametric bootstrap method is the preferred choice in this case. Despite being computationally less efficient, it bypasses the normality assumption that the asymptotic theory requires, while making full use of the small sample size given.

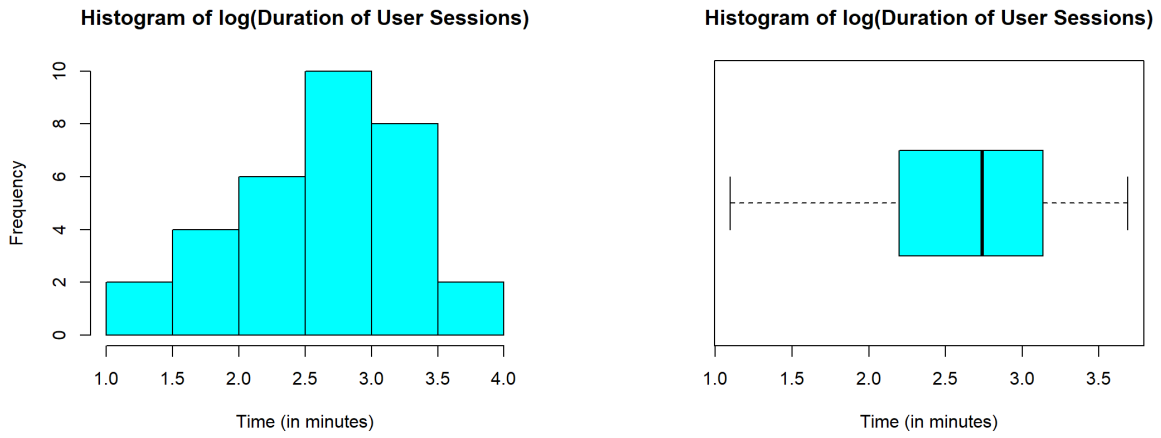


Figure 6: Distribution of log(Duration of User Sessions).

d) The empirical bootstrap sampling distribution of the interquartile ranges is shown in Figure 7. The histogram and the Q-Q plot's distribution is slightly positively skewed. This matches the result of the mean being slightly larger than the median, but still somewhat similar.

Bootstrap 95% confidence interval for the interquartile ranges is (7.75, 18.25).

```
Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
5.50   11.00   13.00   13.05   15.25   21.25
```

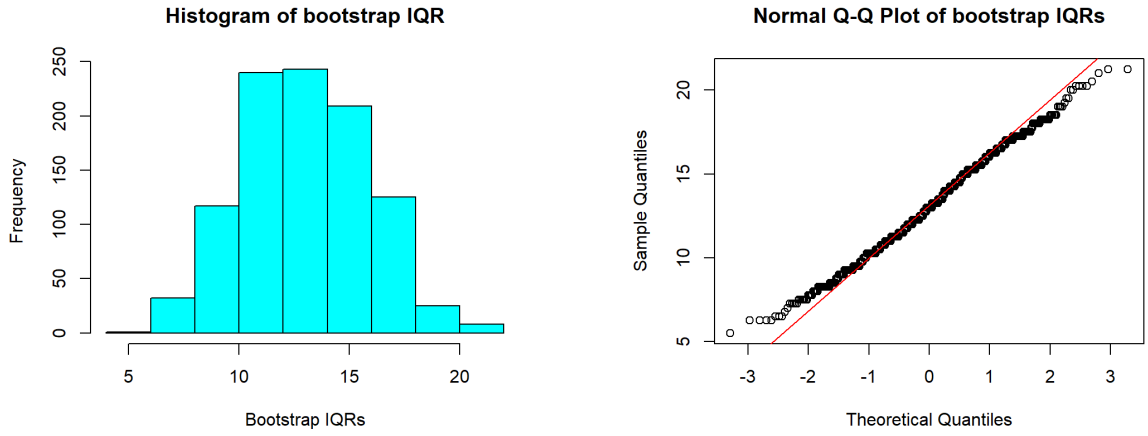


Figure 7: Distribution of bootstrap Interquartile Ranges.

In terms of variability, the 95% confidence interval of the bootstrap IQRs are wider than 95% confidence interval of the bootstrap medians. This suggests that while the middle of the data stays somewhat consistent, the middle spread of the data fluctuates significantly more between each bootstrap sample.

e) Assuming similar assumptions in 2(c), parametric bootstrap was employed to perform a hypothesis test:

$$H_0 : \text{The population median app session duration is more than 15 minutes}$$

$$H_1 : \text{The population median app session duration is 15 minutes or less}$$

Under H_0, the median m_0 needed to be log transformed into $log(15)$.

Figure 8 shows the sampling distribution of the sample median under the H_0 with data coming from $N(log(15), 0.1414405^2)$ distribution with sample median being 2.74 minutes. We find p-value $= 0.8523148$. There is no evidence that the population median app session duration is 15 minutes or less.

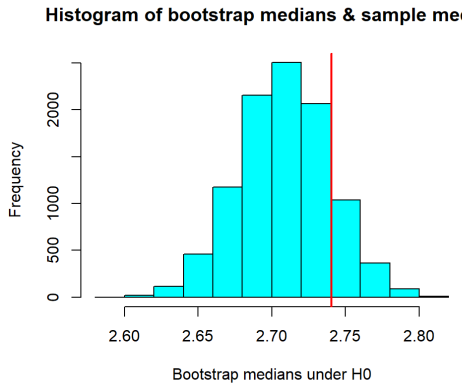


Figure 8: Sampling distribution of bootstrap medians of sample from $N(log(15), 0.1414405^2)$ distribution, with red line showing sample median.

A. Appendix

```r
set.seed(6014)
############################################################
# Question 1
############################################################

# Part a
# load data
file = scan("bug_time.txt")
file

# Histogram
hist(file, main = "Histogram of Bug-fix
Resolution Times", xlab = "Time (in minutes)")

# Boxplot
boxplot(file, horizontal = T, main = "Boxplot of Bug-fix
Resolution Times", xlab = "Time (in minutes)")

# numerical summaries
summary(file)

n = length(file)
n

xbar_bug = mean(file)
xbar_bug

sd_bug = sd(file)
sd_bug

# Part b
# log-transform data
transform_data = log(file)
transform_data

# Compute MLE values
mu_hat = mean(transform_data)
mu_hat

sigma_hat = 1/n * sum((transform_data - mu_hat)^2)
sigma_hat

# Compare with sample mean and variance
# Mean
trans_xbar = mean(transform_data)
trans_xbar

# Standard Deviation
trans_sd = sd(transform_data)
trans_sd

# Variance
trans_var = trans_sd^2
trans_var

# Part c

# Q-Q plot
qqnorm(transform_data, main = "Normal Q-Q Plot of log(Bug-fix
Resolution Times)")
qqline(transform_data, col = "red")

# Histogram of transformed data
hist(transform_data, main = "Histogram of log(Bug-fix
Resolution Times)", xlab = "Time (in minutes)")
```

```r
# Boxplot of transformed data
boxplot(transform_data, horizontal = T, main = "Boxplot of log(Bug-fix
Resolution Times)", xlab = "Time (in minutes)")

# Part d
# Breaking transformed data into 10 cells with equal probability
breaks = qnorm(seq(0,1,by=1/10), mean=mu_hat, sd=sqrt(sigma_hat))
breaks

actual_breaks = exp(breaks)
actual_breaks

# Expectation (e_i > 5)
expectation = n*(1/10)
expectation

# Computing observed frequencies f_i
observed = hist(transform_data, breaks = breaks, plot=F, right=F)$counts
observed

# Calculate test-statistic
Chi2 = sum((observed-expectation)^2/expectation)
Chi2

# Calculate p-value (2 estimated parameters, so 10 - 2- 1 degrees of freedom)
p_value = 1 - pchisq(Chi2,7)
p_value

# Reject H_0, bug-fix resolution times does not follow lognormal distribution.

################################################################
# Question 2
################################################################

# Part a
# load data
file2 = scan("app_session_duration.txt")
file2

# Histogram
hist(file2, main = "Histogram of Duration of User Sessions",
     xlab = "Time (in minutes)")

# Boxplot
boxplot(file2, horizontal = T,
        main = "Boxplot of Histogram of Duration of User Sessions",
        xlab = "Time (in minutes)")

# numerical summaries
summary(file2)

n2 = length(file2)
n2

xbar_app = mean(file2)
xbar_app

sd_app = sd(file2)
sd_app

# Part b
# Sampling
A = 1000
median.a = numeric(A)
for(i in 1:A){ median.a[i] = median(sample(file2, size=32, replace=TRUE)) }

# Summary statistics
summary(median.a)
```

```r
# Histogram
hist(median.a, main="Histogram of bootstrap medians",
     xlab= "Bootstrap medians", col="cyan")

# Boxplot
boxplot(median.a, main="Boxplot of bootstrap medians",
        xlab= "Bootstrap medians", col="cyan", horizontal=TRUE)

# Q-Q plot
qqnorm(median.a, main="Normal Q-Q Plot of bootstrap medians")
qqline(median.a, col = "red")

# 95% CI
bs_CI = quantile(median.a,c(0.025,0.975))
bs_CI

# Part c
# log-transform data
transform_data2 = log(file2)
transform_data2
summary(transform_data2)

# Histogram of transformed data
hist(transform_data2, main="Histogram of log(Duration of User Sessions)",
     xlab= "Time (in minutes)", col="cyan")

# Boxplot of transformed data
boxplot(transform_data2, main="Histogram of log(Duration of User Sessions)",
        xlab= "Time (in minutes)", col="cyan", horizontal=TRUE)

# Estimating parameters using MLE
mu_hat2 = mean(transform_data2)
mu_hat2

sigma_hat2 =  1/n2 * sum((transform_data2 - mu_hat2)^2)
sigma_hat2

# sample median
m_hat = median(transform_data2)
m_hat

fm_hat = 1/(sqrt(2*pi*sigma_hat2))
fm_hat

# Standard error
sem_hat = 1/(2*fm_hat*sqrt(n2))
sem_hat

# 95% CI for log(population median)
log_CI = c(m_hat+qnorm(0.025)*sem_hat,m_hat+qnorm(0.975)*sem_hat)
log_CI

# 95% CI for population median
actual_CI = exp(log_CI)
actual_CI

# Part d
# Sampling
I = 1000
IQR = numeric(I)
for(i in 1:I){

  x = sample(file2, size=32, replace=TRUE)
  IQR[i] = quantile(x,0.75) - quantile(x,0.25)

}
```

```r
# Summary statistics
summary(IQR)

# Histogram
hist(IQR, main="Histogram of bootstrap IQR",
     xlab= "Bootstrap IQRs", col="cyan")

# Boxplot
boxplot(IQR, main="Boxplot of bootstrap IQR",
        xlab= "Bootstrap IQRs", col="cyan", horizontal=TRUE)

# Q-Q plot
qqnorm(IQR, main="Normal Q-Q Plot of bootstrap IQRs")
qqline(IQR, col = "red")

# 95% CI
IQR_CI = quantile(IQR,c(0.025,0.975))
IQR_CI

# sd
sd_IQR = sd(IQR)
sd_IQR

# variance
var_IQR = sd_IQR^2
var_IQR

# Part e
PB = 10000

# log-transform H_0
median.hyp = log(15)
boot.median.H0 = numeric(PB)

# Generating samples
for (i in 1: PB){
  para_boot = rnorm(32, mean = median.hyp, sd = sem_hat) # Notice this is under H0
  boot.median.H0[i] = median(para_boot)
}

# Histogram
hist(boot.median.H0, col="cyan",
     main="Histogram of bootstrap medians & sample median",
     xlab = "Bootstrap medians under H0")
abline(v = m_hat, col="red", lwd=2.0)

# p-value
para_boot.pval = (1 + length(boot.median.H0[boot.median.H0 <= m_hat])) / (PB + 1)
para_boot.pval
```